



University of Asia Pacific

Department of Computer Science & Engineering

Course Title : Microprocessors and Microcontrollers Lab

Course Code : CSE 316

Report No : 03

Report Title : Touch Sensor Door Lock System

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1. Experiment Name

Touch Sensor Door Lock System

2. Objective

To design and implement a touch/push-button based door lock system using Arduino, where a servo motor locks/unlocks the door and a buzzer provides different sound indications for each action.

3. Apparatus / Hardware & Software Requirements

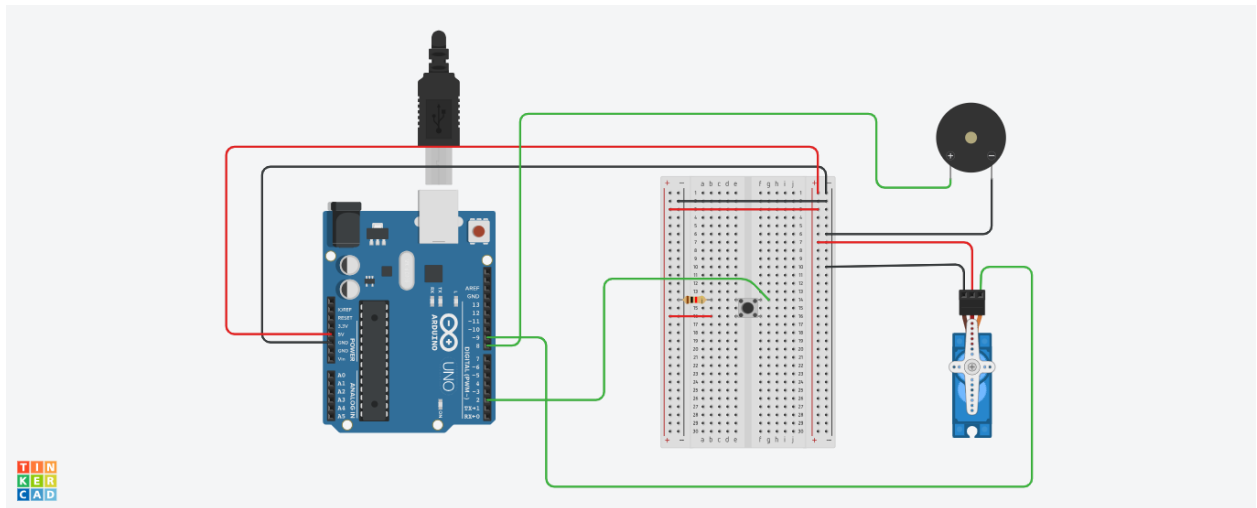
- Hardware

- Arduino Uno
- Push button / Touch sensor (digital input)
- Servo motor
- Buzzer
- Breadboard
- Jumper wires
- USB cable / 5V supply

Software

- Arduino IDE
- Arduino Servo library (`#include <Servo.h>`)

4. Circuit Diagram / Schematic



5. Code / Assembly Program

```
#include <Servo.h>

Servo myServo;

const int PushPin = 2;
const int buzzerPin = 8;
const int servoPin = 9;
bool doorLocked = true;

void setup() {
  pinMode(PushPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  myServo.attach(servoPin);
  myServo.write(0); // Initially
  locked position
}
```

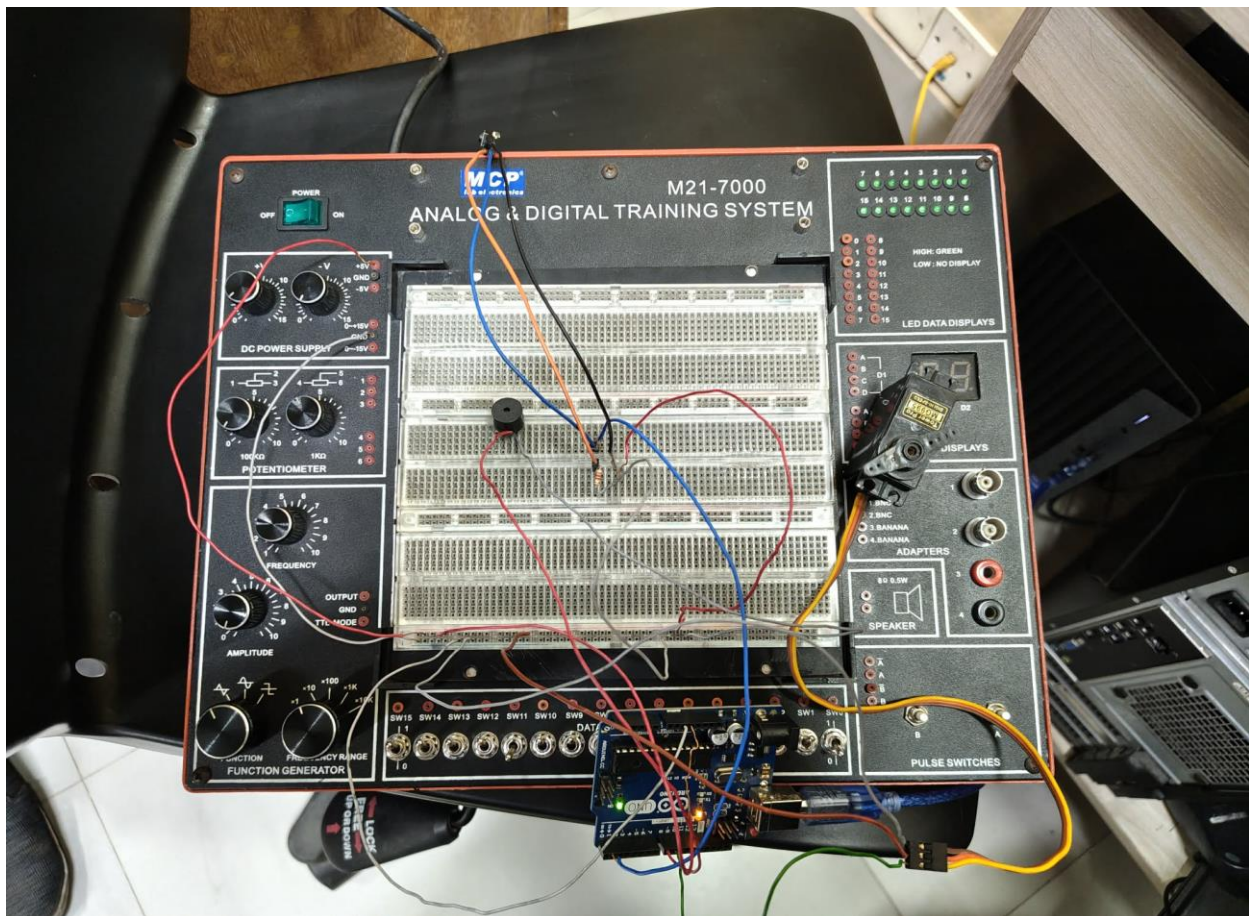
```

}
void loop() {
  int touchState =
  digitalRead(PushPin);
  if (touchState == HIGH) {
    if (doorLocked) {
      // Unlock door
      myServo.write(90);
      tone(buzzerPin, 200, 500000);
      delay(200);
      noTone(buzzerPin);
      doorLocked = false;
    } else {
      // Lock door
      myServo.write(0);
      tone(buzzerPin, 500, 400000);
      delay(400);
      noTone(buzzerPin);
      doorLocked = true;
    }
    delay(1000);
  }
}

```

6. Output / Observations

- The system starts in **locked position** (servo at 0°).
- When the touch/push input is detected **HIGH**, the Arduino toggles the door state:
 - If locked → it unlocks and gives a buzzer tone.
 - If unlocked → it locks and gives a different buzzer tone.
- The system waits briefly (`delay(1000)`) to avoid rapid repeated toggling.



7. Result

After implementing and testing the system, the following observations were recorded:

Unlock Action (when doorLocked = true):

When the button was pressed while the door was locked, the servo rotated to 90° (unlock position). The buzzer produced a low-frequency tone (200 Hz), indicating successful unlocking. The door state changed to unlocked and the system responded correctly.

Lock Action (when doorLocked = false):

When the button was pressed again while the door was unlocked, the servo rotated back to 0° (lock position). The buzzer produced a higher-frequency tone (500 Hz), indicating successful locking. The door state changed back to locked and the cycle repeated properly.

8. Conclusion

This project provided practical experience in building a simple access-control system using Arduino. It strengthened understanding of digital input reading, controlling a servo motor for mechanical movement, and using a buzzer for audio feedback. The experiment also improved skills in implementing toggle logic using a state variable (doorLocked) and coordinating multiple outputs reliably.

Overall, the touch sensor door lock system worked as expected: each press alternated between lock and unlock states with clear visual/mechanical action from the servo and distinct sound feedback from the buzzer. This setup offers a strong foundation for future enhancements such as adding RFID, keypad security, password control, or wireless access features.